

3.1.20**safety function**

function of the machine whose failure can result in an immediate increase of the risk(s)

[SOURCE: ISO 12100:2010, 3.30.]

3.1.21**monitoring**

safety function which ensures that a protective measure is initiated if the ability of a component or an element to perform its function is diminished or if the process conditions are changed in such a way that a decrease of the amount of risk reduction is generated

3.1.22**programmable electronic system****PES**

system for control, protection or monitoring dependent for its operation on one or more programmable electronic devices, including all elements of the system such as power supplies, sensors and other input devices, contactors and other output devices

[SOURCE: IEC 61508-4:1998, 3.3.2, modified.]

3.1.23**performance level****PL**

discrete level used to specify the ability of safety-related parts of control systems to perform a safety function under foreseeable conditions

Note 1 to entry: See [4.5.1](#).

3.1.24**required performance level****PL_r**

performance level (PL) applied in order to achieve the required risk reduction for each safety function

Note 1 to entry: See [Figures 2](#) and [A.1](#).

3.1.25**mean time to dangerous failure****MTTF_D**

expectation of the mean time to dangerous failure

[SOURCE: IEC 62061:2005, 3.2.34, modified.]

3.1.26**diagnostic coverage****DC**

measure of the effectiveness of diagnostics, which may be determined as the ratio between the failure rate of detected dangerous failures and the failure rate of total dangerous failures

Note 1 to entry: Diagnostic coverage can exist for the whole or parts of a safety-related system. For example, diagnostic coverage could exist for sensors and/or logic system and/or final elements.

[SOURCE: IEC 61508-4:1998, 3.8.6, modified.]

3.1.27**protective measure**

measure intended to achieve risk reduction

EXAMPLE 1 Implemented by the designer: inherent design, safeguarding and complementary protective measures, information for use.

EXAMPLE 2 Implemented by the user: organization (safe working procedures, supervision, permit-to-work systems), provision and use of additional safeguards, personal protective equipment, training.

[SOURCE: ISO 12100:2010, 3.19, modified.]

3.1.28

mission time

T_M

period of time covering the intended use of an SRP/CS

3.1.29

test rate

r_t

frequency of automatic tests to detect faults in a SRP/CS, reciprocal value of diagnostic test interval

3.1.30

demand rate

r_D

frequency of demands for a safety-related action of the SRP/CS

3.1.31

repair rate

r_r

reciprocal value of the period of time between detection of a dangerous failure by either an online test or obvious malfunction of the system and the restart of operation after repair or system/component replacement

Note 1 to entry: The repair time does not include the span of time needed for failure-detection.

3.1.32

machine control system

system which responds to input signals from parts of machine elements, operators, external control equipment or any combination of these and generates output signals causing the machine to behave in the intended manner

Note 1 to entry: The machine control system can use any technology or any combination of different technologies (e.g. electrical/electronic, hydraulic, pneumatic, mechanical).

3.1.33

safety integrity level

SIL

discrete level (one out of a possible four) for specifying the safety integrity requirements of the safety functions to be allocated to the E/E/PE safety-related systems, where safety integrity level 4 has the highest level of safety integrity and safety integrity level 1 has the lowest

[SOURCE: IEC 61508-4:1998, 3.5.6.]

3.1.34

limited variability language

LVL

type of language that provides the capability of combining predefined, application-specific library functions to implement the safety requirements specifications

Note 1 to entry: Typical examples of LVL (ladder logic, function block diagram) are given in IEC 61131-3.

Note 2 to entry: A typical example of a system using LVL: PLC.

[SOURCE: IEC 61511-1:2003, 3.2.80.1.2, modified.]

3.1.35**full variability language****FVL**

type of language that provides the capability of implementing a wide variety of functions and applications

EXAMPLE C, C++, Assembler.

Note 1 to entry: A typical example of systems using FVL: embedded systems.

Note 2 to entry: In the field of machinery, FVL is found in embedded software and rarely in application software.

[SOURCE: IEC 61511-1:2003, 3.2.80.1.3, modified.]

3.1.36**application software**

software specific to the application, implemented by the machine manufacturer, and generally containing logic sequences, limits and expressions that control the appropriate inputs, outputs, calculations and decisions necessary to meet the SRP/CS requirements

3.1.37**embedded software**

firmware

system software

software that is part of the system supplied by the control manufacturer and which is not accessible for modification by the user of the machinery

Note 1 to entry: Embedded software is usually written in FVL.

3.1.38**high demand or continuous mode**

mode of operation in which the frequency of demands on a SRP/CS is greater than one per year or the safety related control function retains the machine in a safe state as part of normal operation

[SOURCE: IEC 62061:2012, 3.2.27, modified.]

3.1.39**proven in use**

demonstration, based on an analysis of operational experience for a specific configuration of an element, that the likelihood of dangerous systematic faults is low enough so that every safety function that uses the element achieves its required performance level (PL_r)

[SOURCE: IEC 61508-4:2010, 3.8.18, modified.]

3.2 Symbols and abbreviated terms

See [Table 1](#).

Table 1 — Symbols and abbreviated terms

Symbol or abbreviation	Description	Definition or occurrence
a, b, c, d, e	Denotation of performance levels	Table 3
AOPD	Active optoelectronic protective device (e.g. light barrier)	Annex H
B, 1, 2, 3, 4	Denotation of categories	Table 7
B_{10D}	Number of cycles until 10 % of the components fail dangerously (for pneumatic and electromechanical components)	Annex C
Cat.	Category	3.1.2
CC	Current converter	Annex I
CCF	Common cause failure	3.1.6
DC	Diagnostic coverage	3.1.26
DC_{avg}	Average diagnostic coverage	E.2
F, F1, F2	Frequency and/or time of exposure to the hazard	A.2.2
FB	Function block	4.6.3
FVL	Full variability language	3.1.35
FMEA	Failure modes and effects analysis	7.2
I, I1, I2	Input device, e.g. sensor	6.2
i, j	Index for counting	Annex D
I/O	Inputs/outputs	Table E.1
i_{ab}, i_{bc}	Interconnecting means	Figure 4
K1A, K1B	Contactors	Annex I
L, L1, L2	Logic	6.2
LVL	Limited variability language	3.1.34
M	Motor	Annex I
MTTF	Mean time to failure	Annex C
$MTTF_D$	Mean time to dangerous failure	3.1.25
$n, N, \tilde{N}$	Number of items	6.3, D.1
N_{low}	Number of SRP/CS with PL_{low} in a combination of SRP/CS	6.3
n_{op}	Mean number of annual operations	Annex C
O, O1, O2, OTE	Output device, e.g. actuator	6.2
P, P1, P2	Possibility of avoiding the hazard	A.2.3
PES	Programmable electronic system	3.1.22
PFH_D	average probability of dangerous failure per hour	Table 3 and Table K.1
PL	Performance level	3.1.23
PLC	Programmable logic controller	Annex I
PL_{low}	Lowest performance level of a SRP/CS in a combination of SRP/CS	6.3
PL_r	Required performance level	3.1.24
r_D	Demand rate	3.1.30
r_t	Test rate	3.1.29
RS	Rotation sensor	Annex I
S, S1, S2	Severity of injury	A.2.1
SW1A, SW1B, SW2	Position switches	Annex I

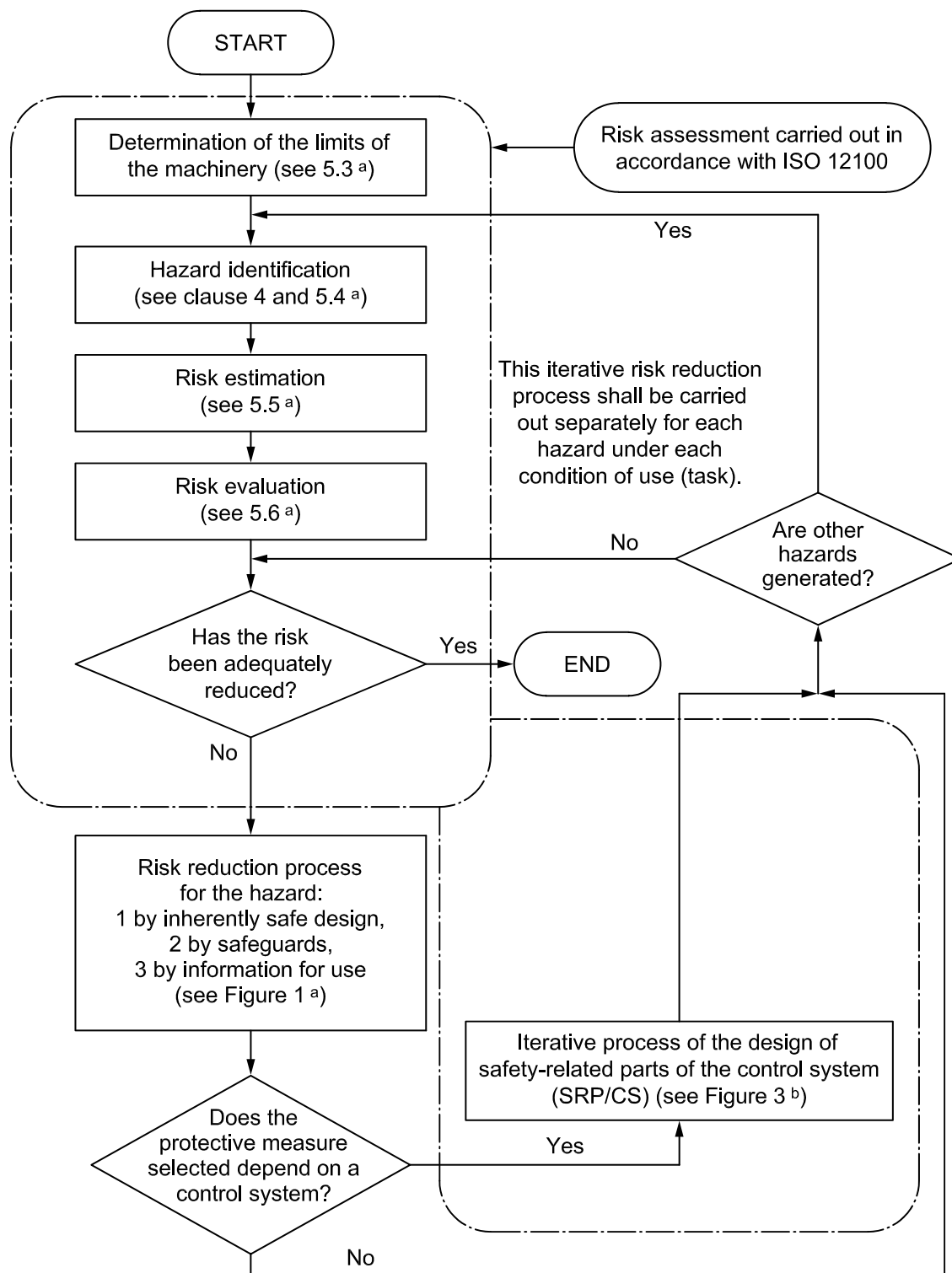
Table 1 (continued)

Symbol or abbreviation	Description	Definition or occurrence
SIL	Safety integrity level	Table 4
SRASW	Safety-related application software	4.6.3
SRESW	Safety-related embedded software	4.6.2
SRP	Safety-related part	General
SRP/CS	Safety-related part of a control system	3.1.1
TE	Test equipment	6.2
T_M	Mission time	3.1.28
T_{10D}	Mean time until 10 % of the components fail dangerously	Annex C

4 Design considerations

4.1 Safety objectives in design

The SRP/CS shall be designed and constructed so that the principles of ISO 12100 are fully taken into account (see [Figures 1](#) and [3](#)). All intended use and reasonable foreseeable misuse shall be considered.



a Refers to ISO 12100:2010

b Refers to this part of ISO 13849

Figure 1 — Overview of risk assessment/risk reduction

4.2 Strategy for risk reduction

4.2.1 General

The strategy for risk reduction at the machine is given in ISO 12100:2010, 6.1, and further guidance is given in ISO 12100:2010, 6.2 (inherent design measures) and 6.3 (safeguarding and complementary protective measures). This strategy covers the whole life cycle of the machine.

The hazard analysis and risk reduction process for a machine requires that hazards are eliminated or reduced through a hierarchy of measures:

- hazard elimination or risk reduction by design (see ISO 12100:2010, 6.2);
- risk reduction by safeguarding and possibly complementary protective measures (see ISO 12100:2010, 6.3);
- risk reduction by the provision of information for use about the residual risk (see ISO 12100:2010, 6.4).

4.2.2 Contribution to the risk reduction by the control system

The purpose in following the overall design procedure for the machine is to achieve the safety objectives (see 4.1). The design of the SRP/CS to provide the required risk reduction is an integral subset of the overall design procedure for the machine. The SRP/CS provides safety function(s) at a PL which achieves the required risk reduction. In providing safety function(s), either as an inherently safe part of the design or as a control for an interlocking guard or protective device, the design of the SRP/CS is a part of the strategy for risk reduction. This is an iterative process and is illustrated in Figures 1 and 3.

NOTE There is no need to apply this strategy of risk reduction on non-safety related parts of control systems or purely functional elements of a machine (see ISO/TR 22100-2:2013, Clause 3).

For each safety function, the characteristics (see Clause 5) and the required performance level shall be specified and documented in the safety requirements specification.

In this part of ISO 13849 the performance levels are defined in terms of probability of dangerous failure per hour. Five performance levels are set out, from the lowest PL a to the highest PL e with defined ranges of probability of a dangerous failure per hour (see Table 2).

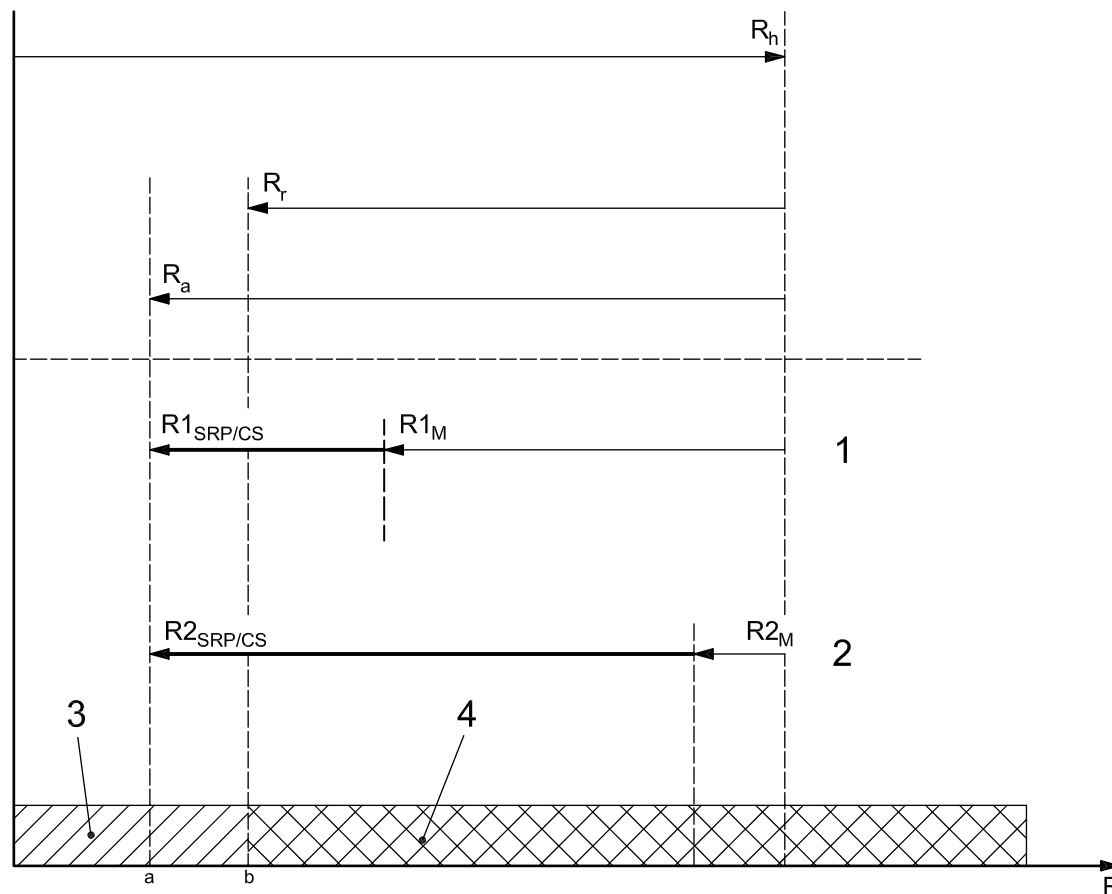
In order to achieve a PL, beside quantifiable aspects, it is also necessary to satisfy requirements related to qualitative aspects of PL (see 4.5).

Table 2 — Performance levels (PL)

PL	Average probability of dangerous failure per hour (PFH _D) 1/h
a	$\geq 10^{-5}$ to $< 10^{-4}$
b	$\geq 3 \times 10^{-6}$ to $< 10^{-5}$
c	$\geq 10^{-6}$ to $< 3 \times 10^{-6}$
d	$\geq 10^{-7}$ to $< 10^{-6}$
e	$\geq 10^{-8}$ to $< 10^{-7}$

From the risk assessment (see ISO 12100) at the machine, the designer shall decide the contribution to the reduction of risk which needs to be provided by each relevant safety function which is carried out by the SRP/CS(s). This contribution does not cover the overall risk of the machinery under control, e.g. not the overall risk of a mechanical press, or washing machine is considered, but that part of risk reduced by the application of particular safety functions. Examples of such functions are the stopping function initiated by using an electro-sensitive protective device on a press or the door-locking function of a washing machine.

Risk reduction can be achieved by applying various protective measures (both SRP/CS and non SRP/CS) with the end result of achieving a safe condition (see [Figure 2](#)).



Key

R_h	for a specific hazardous situation, the risk before protective measures are applied
R_r	risk reduction required from protective measures
R_a	actual risk reduction achieved with protective measures
1	Solution 1 — important part of risk reduction due to protective measures other than SRP/CS (e.g. mechanical measures), small part of risk reduction due to SRP/CS
2	solution 2 — important part of risk reduction due to the SRP/CS (e.g. light curtain), small part of risk reduction due to protective measures other than SRP/CS (e.g. mechanical measures)
3	adequately reduced risk
4	inadequately reduced risk
R	risk
a	residual risk obtained by solutions 1 and 2
b	adequately reduced risk
$R1_{SRP/CS}$	risk reduction from the safety function carried out by the SRP/CS
$R2_{SRP/CS}$	risk reduction from the safety function carried out by the SRP/CS
$R1_M, R2_M$	risk reduction from protective measures other than SRP/CS (e.g. mechanical measures)

NOTE See ISO 12100 for further information on risk reduction.

Figure 2 — Overview of the risk reduction process for each hazardous situation